

Case Study: Engel Curves and Quantile Regression

Workshop Exercise

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1 Learning Objectives

By the end of this exercise you will be able to:

- Fit quantile regression models using the Engel dataset
- Compare OLS and quantile regression estimates
- Examine how the relationship between income and food expenditure varies across quantiles
- Construct confidence intervals using `se = "nid"`
- Interpret a coefficient process plot
- Write up quantile regression findings in plain language

2 Introduction

A classic question in economics is:

How does household food expenditure change as income increases?

The Engel dataset is one of the most famous examples used in the quantile regression literature and is included in the `quantreg` package.

Unlike the CPS1988 wage example, this dataset contains only two variables:

- Household income
- Household food expenditure

This allows us to focus entirely on understanding quantile regression.

2.1 Exercise 1: Exploring the Data

Inspect the dataset

```
glimpse(engel)
```

Rows: 235

Columns: 2

```
$ income <dbl> 420.1577, 541.4117, 901.1575, 639.0802, 750.8756, 945.7989, 82~
```

```
$ foodexp <dbl> 255.8394, 310.9587, 485.6800, 402.9974, 495.5608, 633.7978, 63~
```

```
summary(engel)
```

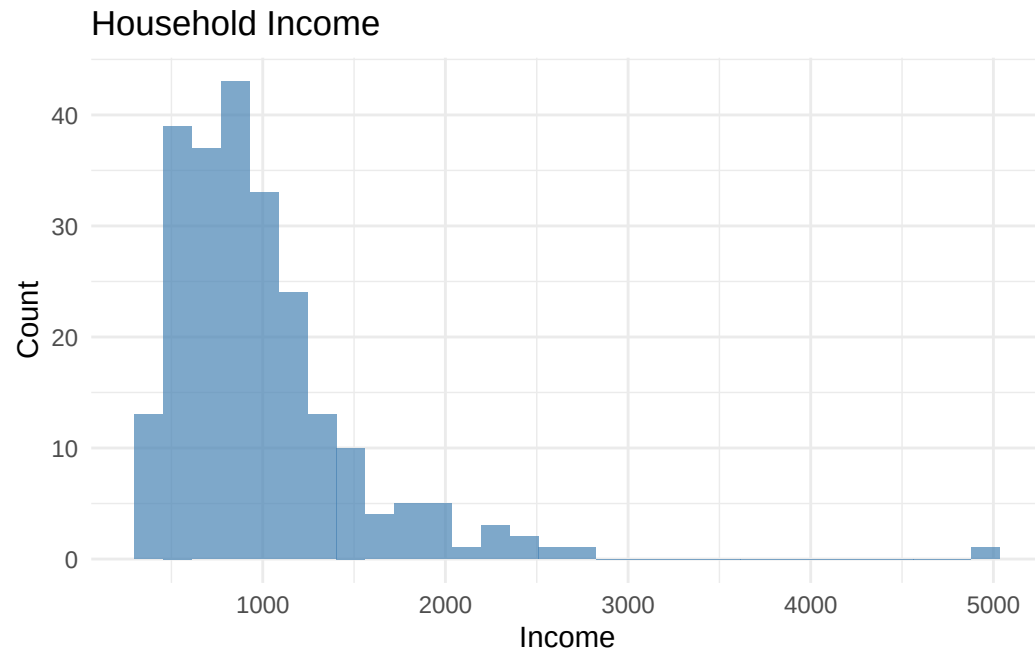
income	foodexp
Min. : 377.1	Min. : 242.3
1st Qu.: 638.9	1st Qu.: 429.7
Median : 884.0	Median : 582.5
Mean : 982.5	Mean : 624.2
3rd Qu.: 1164.0	3rd Qu.: 743.9
Max. : 4957.8	Max. : 2032.7

Questions

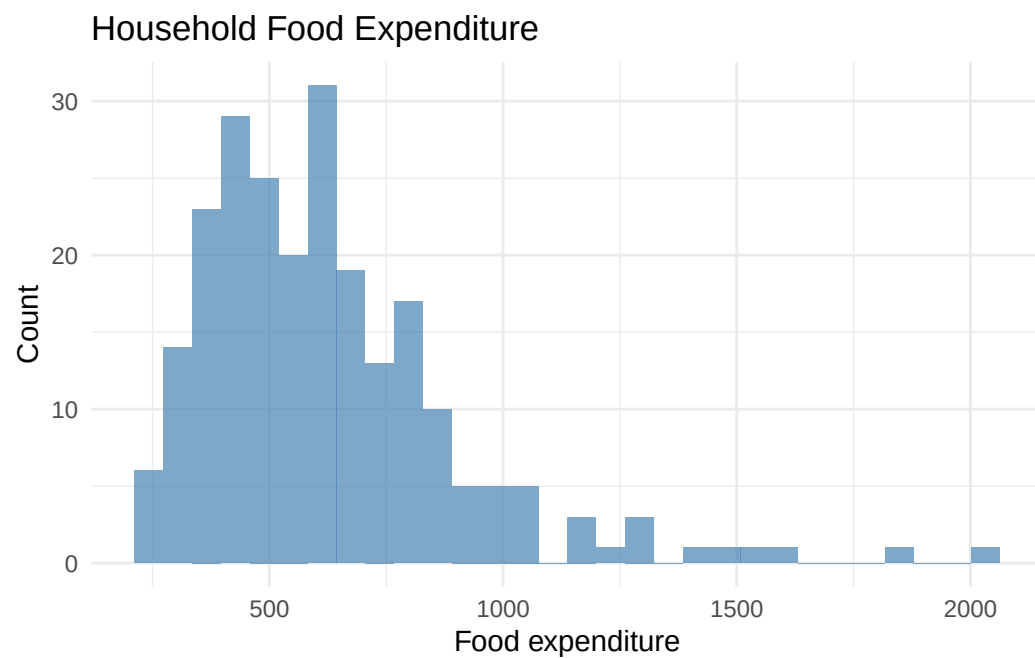
1. How many households are included?
 2. Which variable is the outcome?
 3. Which variable is the predictor?
-

2.2 Visualising the Variables

```
ggplot(engel, aes(x = income)) +  
  geom_histogram(bins = 30,  
                 fill = "steelblue",  
                 alpha = 0.7) +  
  labs(  
    title = "Household Income",  
    x = "Income",  
    y = "Count"  
  )
```



```
ggplot(engel, aes(x = foodexp)) +  
  geom_histogram(bins = 30,  
                 fill = "steelblue",  
                 alpha = 0.7) +  
  labs(  
    title = "Household Food Expenditure",  
    x = "Food expenditure",  
    y = "Count"  
  )
```

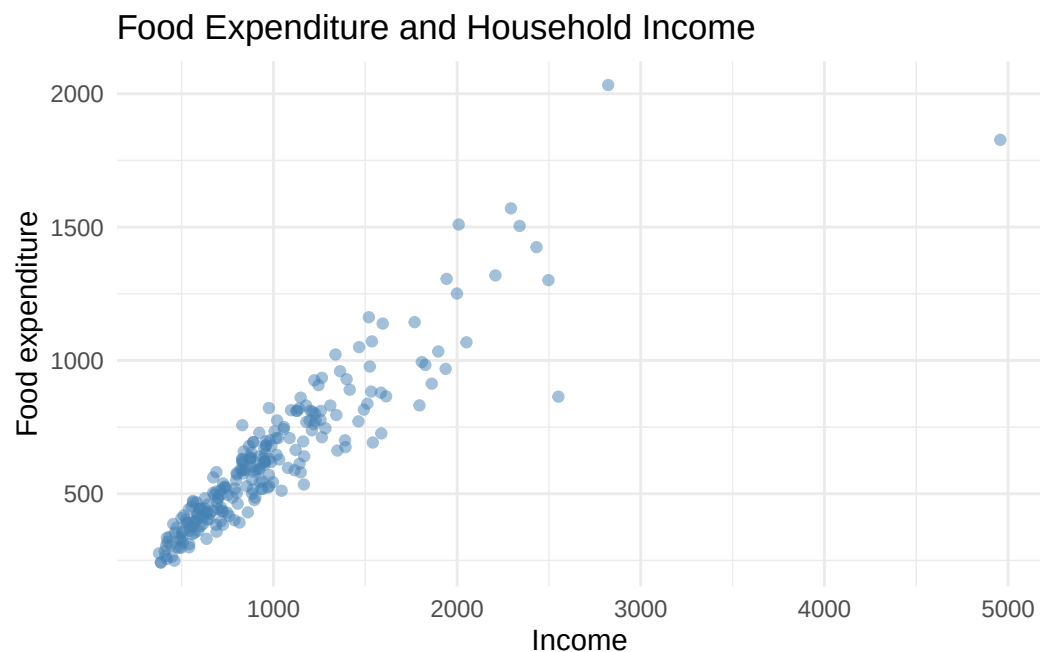


Questions

1. Is income skewed?
2. Is food expenditure skewed?
3. Why might averages be insufficient to describe these distributions?

2.3 Exercise 2: Income and Food Expenditure

```
ggplot(  
  engel,  
  aes(  
    x = income,  
    y = foodexp  
  )  
) +  
  geom_point(alpha = 0.5,  
             colour = "steelblue") +  
  labs(  
    title = "Food Expenditure and Household Income",  
    x = "Income",  
    y = "Food expenditure"  
  )  
)
```



Questions

1. Is the relationship positive?
2. Does variability appear constant across the income range?
3. Why might quantile regression be useful here?

2.4 Exercise 3: OLS Baseline

```
ols_fit <- lm(  
  foodexp ~ income,  
  data = engel  
)
```

```
)  
  
summary(ols_fit)
```

Call:

```
lm(formula = foodexp ~ income, data = engel)
```

Residuals:

```
      Min       1Q   Median       3Q      Max  
-725.70  -60.24   -4.32   53.41  515.77
```

Coefficients:

```
              Estimate Std. Error t value Pr(>|t|)  
(Intercept) 147.47539   15.95708    9.242  <2e-16 ***  
income       0.48518    0.01437   33.772  <2e-16 ***  
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 114.1 on 233 degrees of freedom

Multiple R-squared: 0.8304, Adjusted R-squared: 0.8296

F-statistic: 1141 on 1 and 233 DF, p-value: < 2.2e-16

Questions

1. What is the coefficient on income?
2. Interpret the coefficient.
3. What does OLS tell us about the relationship between income and food expenditure?

2.5 Exercise 4: Median Regression

```
fit_median <- rq(  
  foodexp ~ income,  
  tau = 0.50,  
  data = engel  
)  
  
summary(  
  fit_median,  
  se = "nid"  
)
```

Call: rq(formula = foodexp ~ income, tau = 0.5, data = engel)

tau: [1] 0.5

Coefficients:

```
              Value    Std. Error t value  Pr(>|t|)  
(Intercept) 81.48225  19.25066    4.23270  0.00003  
income       0.56018   0.02828   19.81032  0.00000
```

```
# Alternative:  
# summary(  
#   fit_median,
```

```
# se = "boot",
# R = 1000
# )
```

Questions

1. How does the median regression estimate compare with OLS?
2. Why might the two estimates differ?
3. What does median regression estimate that OLS does not?

2.6 Exercise 5: Multiple Quantiles

```
taus <- c(
  0.10,
  0.25,
  0.50,
  0.75,
  0.90
)

fit_multi <- rq(
  foodexp ~ income,
  tau = taus,
  data = engel
)

coef(fit_multi)
```

```

              tau= 0.10  tau= 0.25  tau= 0.50  tau= 0.75  tau= 0.90
(Intercept) 110.1415742 95.4835396 81.4822474 62.3965855 67.3508721
income       0.4017658  0.4741032  0.5601806  0.6440141  0.6862995
```

Questions

1. Is the income coefficient constant across quantiles?
2. Which quantile has the largest coefficient?
3. What does this imply about spending behaviour?

2.6.1 Formal Test of Equality Across Quantiles

```
anova(fit_multi)
```

Quantile Regression Analysis of Deviance Table

Model: foodexp ~ income

Joint Test of Equality of Slopes: tau in { 0.1 0.25 0.5 0.75 0.9 }

```

Df Resid Df F value    Pr(>F)
1  4      1171  11.466 3.927e-09 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Questions

4. What is the null hypothesis being tested?
5. What does the p-value suggest?

6. Is there evidence that the income effect differs across quantiles?

2.7 Exercise 6: Confidence Intervals

```
income_df <- map_dfr(taus, \(t){

  fit <- rq(
    foodexp ~ income,
    tau = t,
    data = engel
  )

  s <- summary(
    fit,
    se = "nid"
  )

  # Alternatives:
  # s <- summary(fit, se = "boot", R = 50)    # quick experimentation
  # s <- summary(fit, se = "boot", R = 1000)  # more stable bootstrap inference

  cm <- s$coefficients

  tibble(
    tau    = t,
    est    = cm["income", "Value"],
    lower  = cm["income", "Value"] -
      1.96 * cm["income", "Std. Error"],
    upper  = cm["income", "Value"] +
      1.96 * cm["income", "Std. Error"]
  )
})

income_df
```

```
# A tibble: 5 x 4
  tau    est lower upper
<dbl> <dbl> <dbl> <dbl>
1  0.1  0.402 0.323 0.481
2  0.25 0.474 0.417 0.531
3  0.5   0.560 0.505 0.616
4  0.75 0.644 0.598 0.690
5  0.9   0.686 0.630 0.742
```

Questions

1. Are all estimated income effects positive?
2. How do the confidence intervals change across quantiles?
3. Do the intervals suggest that the effect of income is constant?
4. Why are confidence intervals useful when interpreting quantile regression results?

2.8 Exercise 7: Coefficient Plot

```

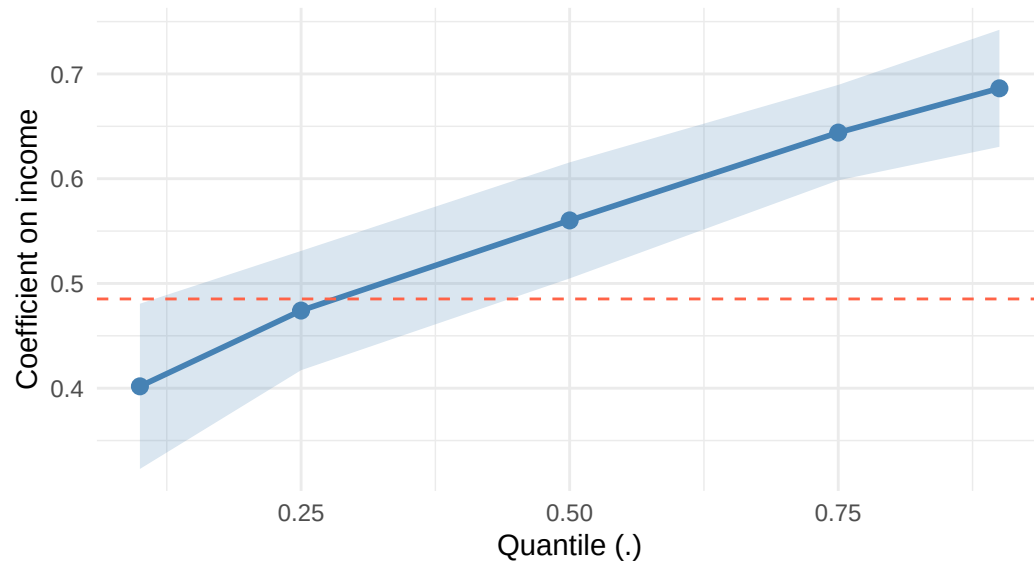
ols_income <- coef(ols_fit)["income"]

ggplot(
  income_df,
  aes(
    x = tau,
    y = est
  )
) +
  geom_ribbon(
    aes(
      ymin = lower,
      ymax = upper
    ),
    alpha = 0.20,
    fill = "steelblue"
  ) +
  geom_line(
    colour = "steelblue",
    linewidth = 1
  ) +
  geom_point(
    colour = "steelblue",
    size = 2.5
  ) +
  geom_hline(
    yintercept = ols_income,
    linetype = "dashed",
    colour = "tomato"
  ) +
  labs(
    title = "Income Effect Across Quantiles",
    subtitle = "Blue = Quantile Regression · Red dashed = OLS",
    x = "Quantile ( )",
    y = "Coefficient on income"
  )

```


Income Effect Across Quantiles

Blue = Quantile Regression · Red dashed = OLS



Questions

1. What pattern do you observe?
2. Does the effect of income appear constant?
3. Does OLS adequately summarise the relationship?

2.9 Exercise 8: Interpretation

Complete the following paragraph.

At the 10th quantile, an additional unit of income is associated with approximately _____ additional units of food expenditure. At the 90th quantile, the corresponding estimate is _____. This suggests that _____. The OLS estimate of _____ provides _____.

Reflection

1. What did quantile regression reveal that OLS could not?
2. Which result was most surprising?
3. In what other applications might quantile regression be useful?